

2026光通信技术课堂板书

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Google

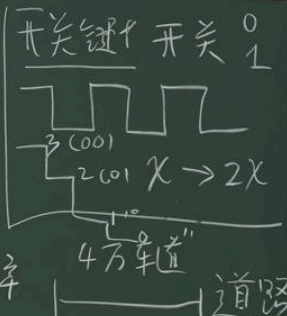
NotebookLM

上传pdf

问答

知识点
问题
考试

开关0
1



1 MHz

bits

→

Mbps

电压调制 × b = 数据速率

骨干网

频谱资源

$\frac{80nm}{0.8nm} \times 100GHz = 10^4GHz = 10^7MHz$

频谱效率

★

数据速率

$\frac{bit/s}{Hz} = \frac{bit/s}{Hz}$

4车道

大巴 → 双双层巴士

车道

1525-1565

带宽(频谱宽度)

⇒

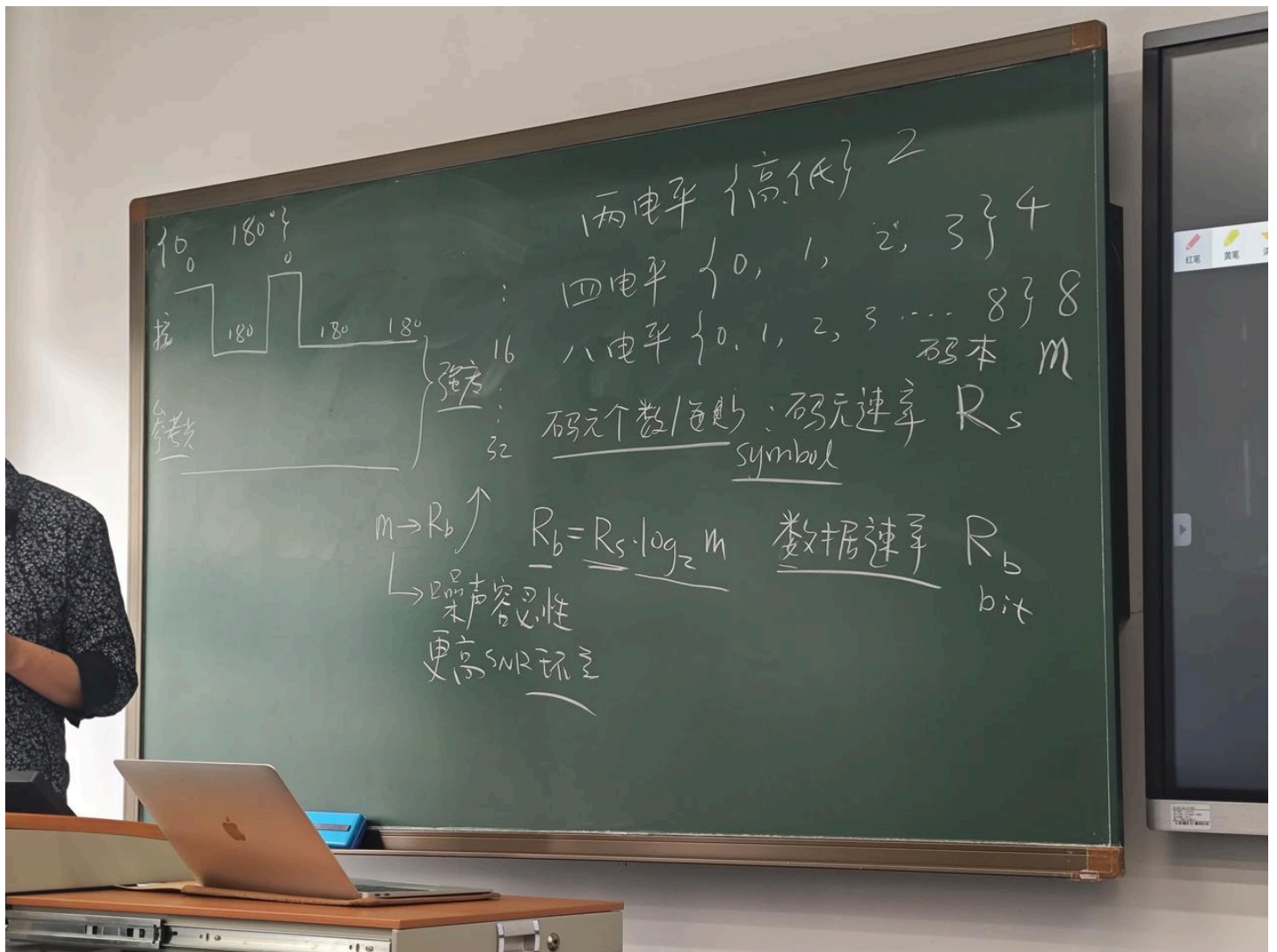
骨干/城域

C波段

光纤通信

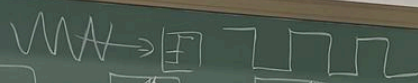


m ↑ Rb ↑ → 距离 ↓



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Tx 发送



1MHz

判决

1MHz = 1ns

采样 1MHz

DSP

Sin wt

cos wt

时钟恢复

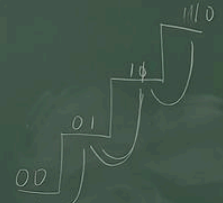
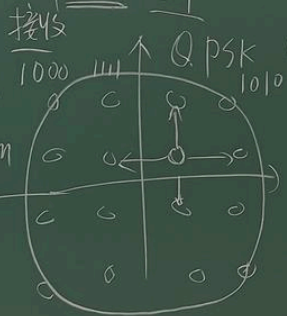
(DSP / 锁相环)

1525-1565nm

40nm

0.8nm x 100GHz

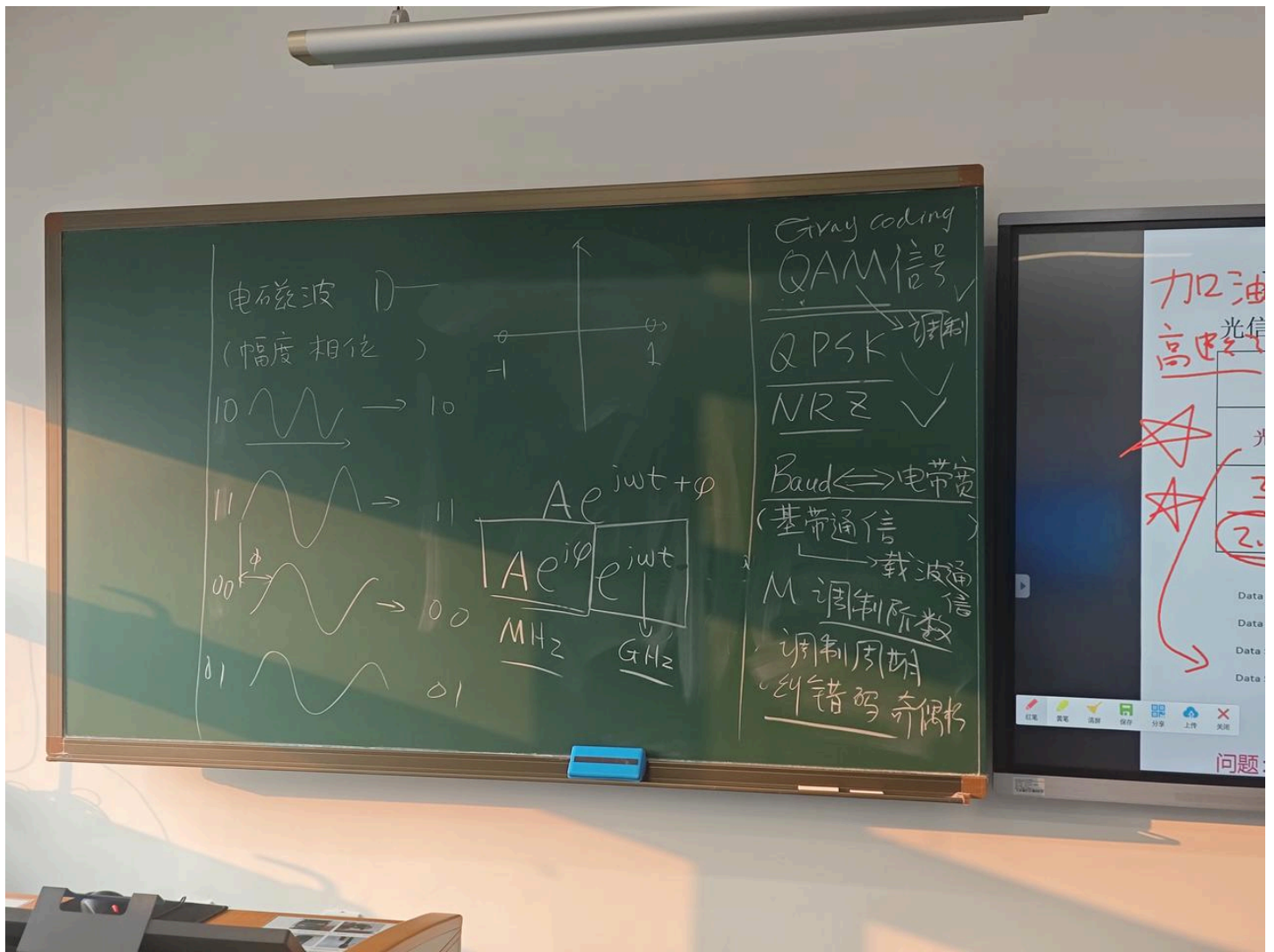
5000 GHz



$$\sin wt \cos wt = 0 \quad \text{if } Q \in \{1, -1\}$$

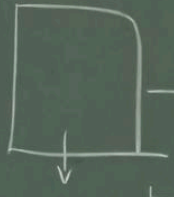
$$I \cdot \cos wt + Q \cdot \sin wt$$

$$I \cdot Q \in \{-3, -1, 1, 3\}$$



0420

$x(t)$



$x(t)$

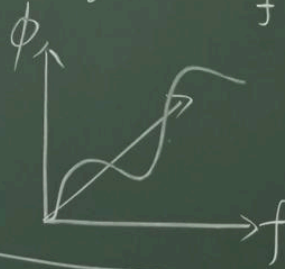
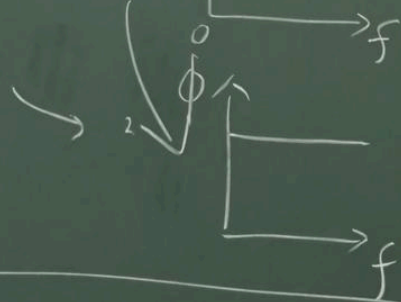
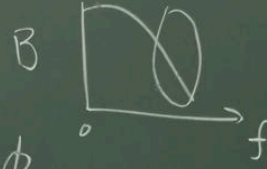
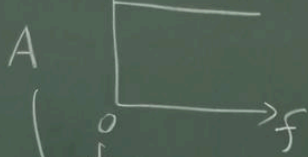
$(t - \tau)$



$x(t)$

幅频

相频



信号 频率 幅度
探测器 放大器

绝对功率



$$\text{dBm}_2 = \text{dBm}_1 + \text{dB}_x$$

证明

$$m = 10 \log_{10} \frac{P_2}{P_1} \Rightarrow$$

$$\text{dB}_x = \text{dBm}_2 - \text{dBm}_1,$$

$$\text{dBm}_2 = \text{dBm}_1 + \text{dB}_x$$

信号频率幅度
探测器放大器

绝对功率



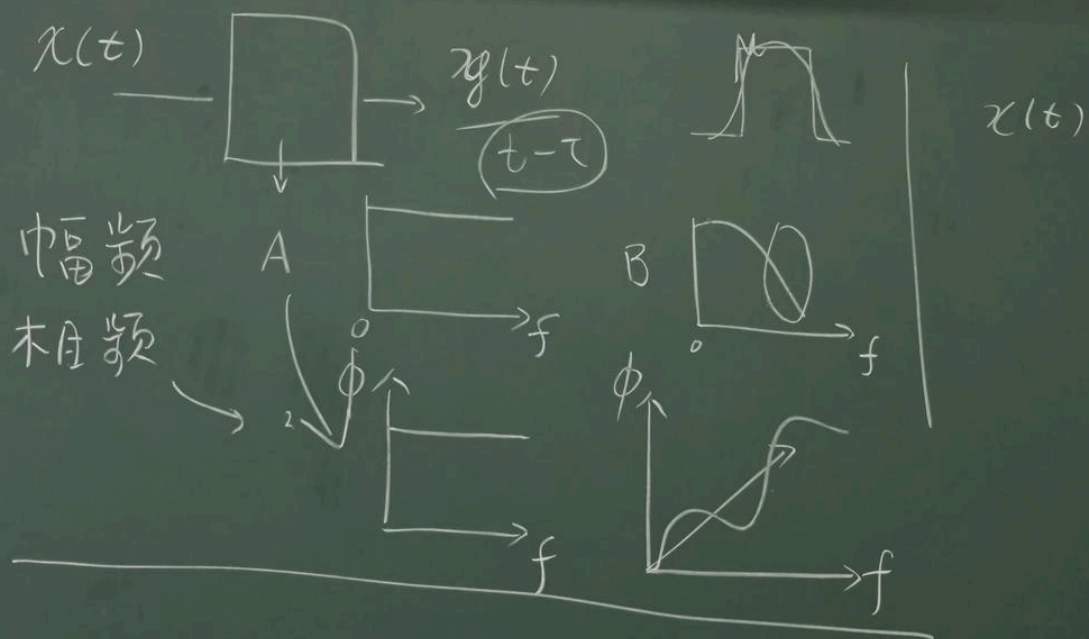
$$\text{dBm}_2 = \text{dBm}_1 + \text{dB}_x$$

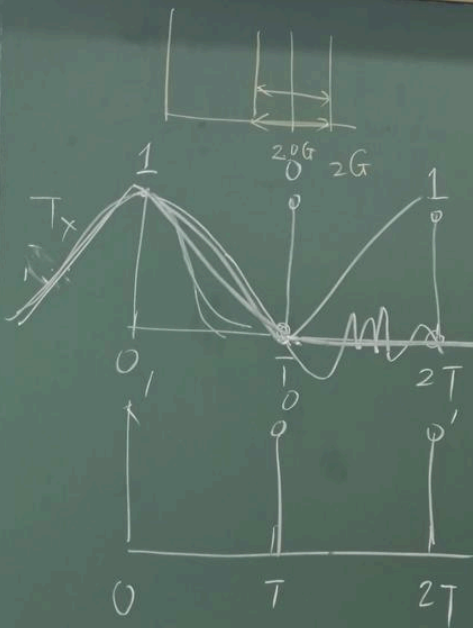
证明

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{P_2}{P_1} \Rightarrow$$

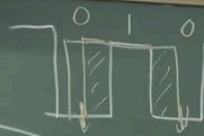
$$\text{dB}_x = \text{dBm}_2 - \text{dBm}_1,$$

$$\text{dBm}_2 = \text{dBm}_1 + \text{dB}_x$$





无码间干扰

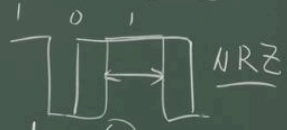


$$\Delta\lambda \cdot L \cdot M \ll \frac{T}{L \Delta\lambda}$$

$$= 2 \times 10 \times 110$$

$$= 2200 \text{ ps} = 2.2 \text{ ns}$$

LED 光源 BH2



10nm
20nm



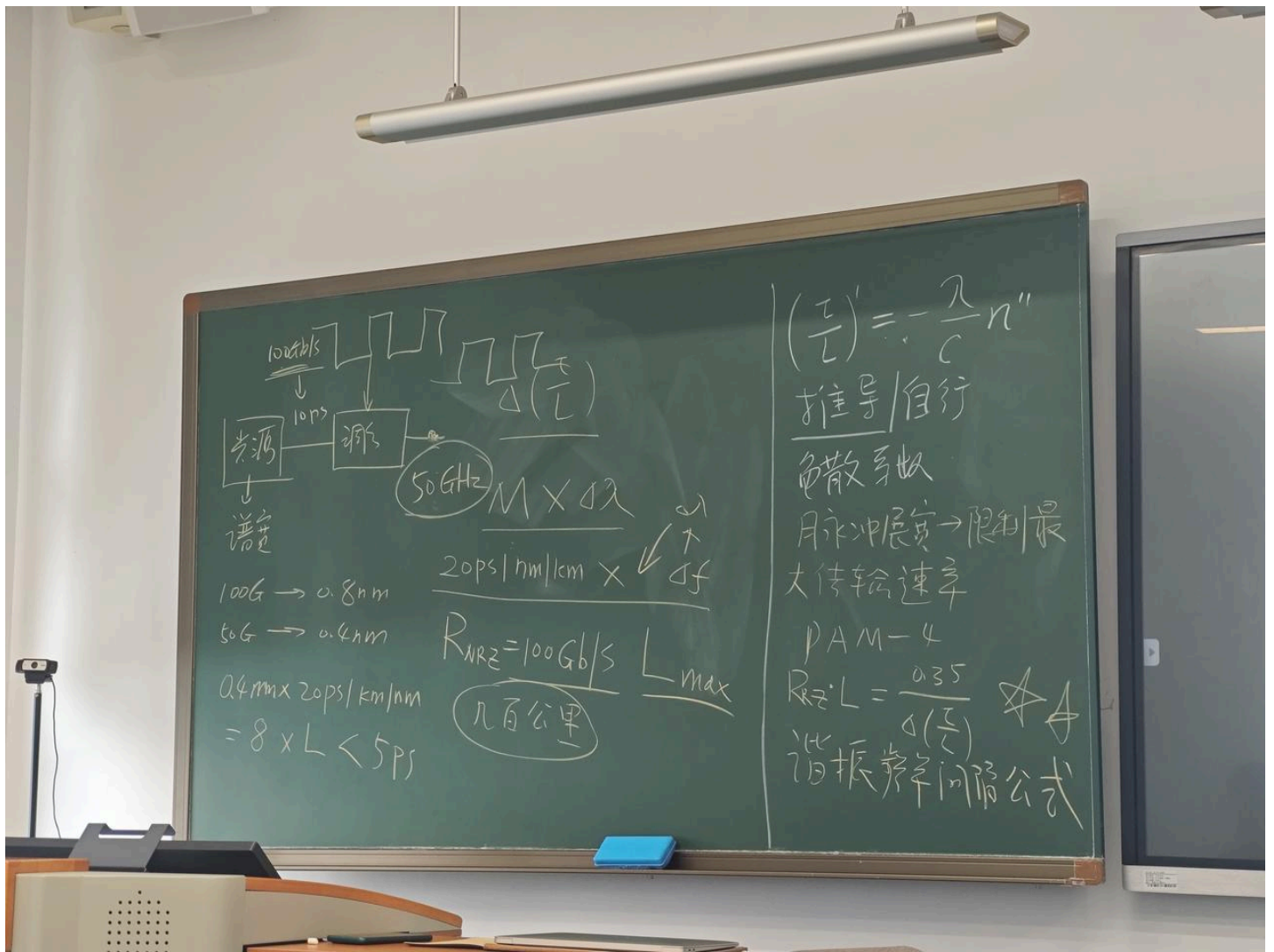
Δλ = 2nm

L = 10km

$$\Delta T = \frac{1}{2.2 \text{ ns}} \left(T \right) > 2.2 \text{ ns}$$

$$10^{-9}$$

$$0.82 \mu\text{m} \rightarrow \lambda$$

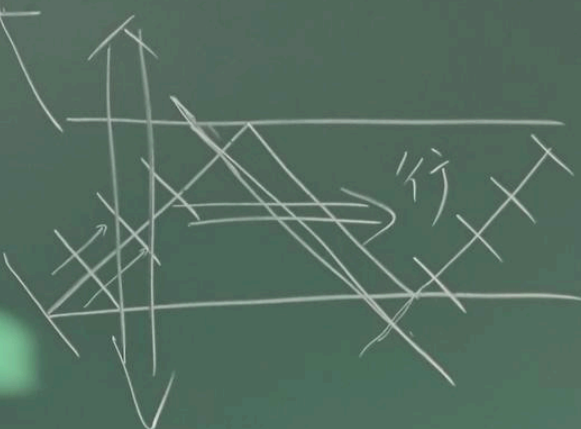


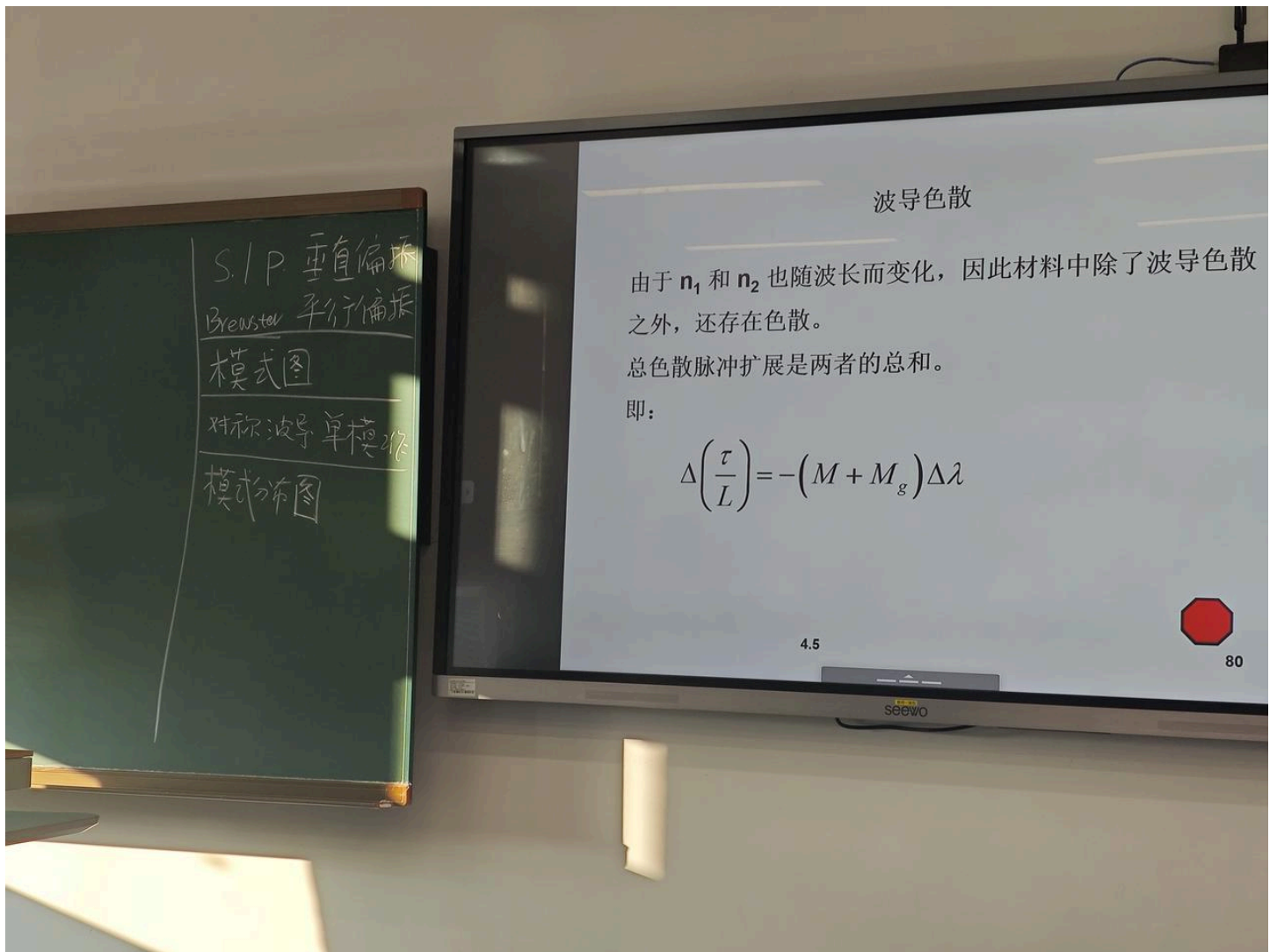


FC/PC 插拔

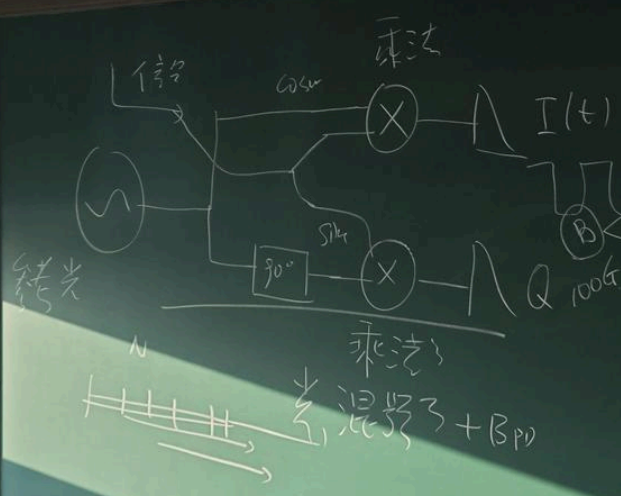
$$\underline{A(x,y) e^{-i k z}} \quad \underline{k(\omega)}$$

$$\frac{d\omega}{d\beta}$$





0507



模式材料
波导
光源
调制
主导因素

$M \cdot \Delta \lambda$
 $\omega (200 THz)$

波分+模式

3dB 耦合器

光调制 $\rightarrow$ 频率啁啾

Q1XK 光调制原理

16QPSK 光调制

$$A(x, y) e^{i\beta(\omega)z}$$

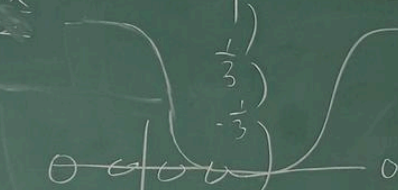
AT 光互联 $\{\pm 1 \pm \frac{1}{3}\} + j\{\pm 1 \pm \frac{1}{3}\}$ QPSK

多模光纤

四电平

$\Delta f \rightarrow \Delta t$

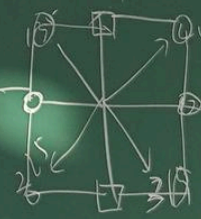
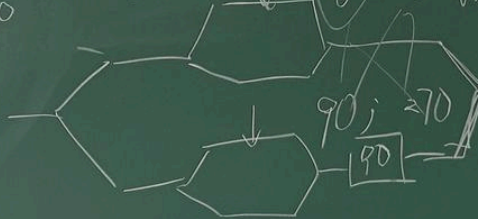
$$V(t) = \pm \frac{V_R}{2} + \frac{V_R}{2} \left\{ V_R, 0 \right\}$$

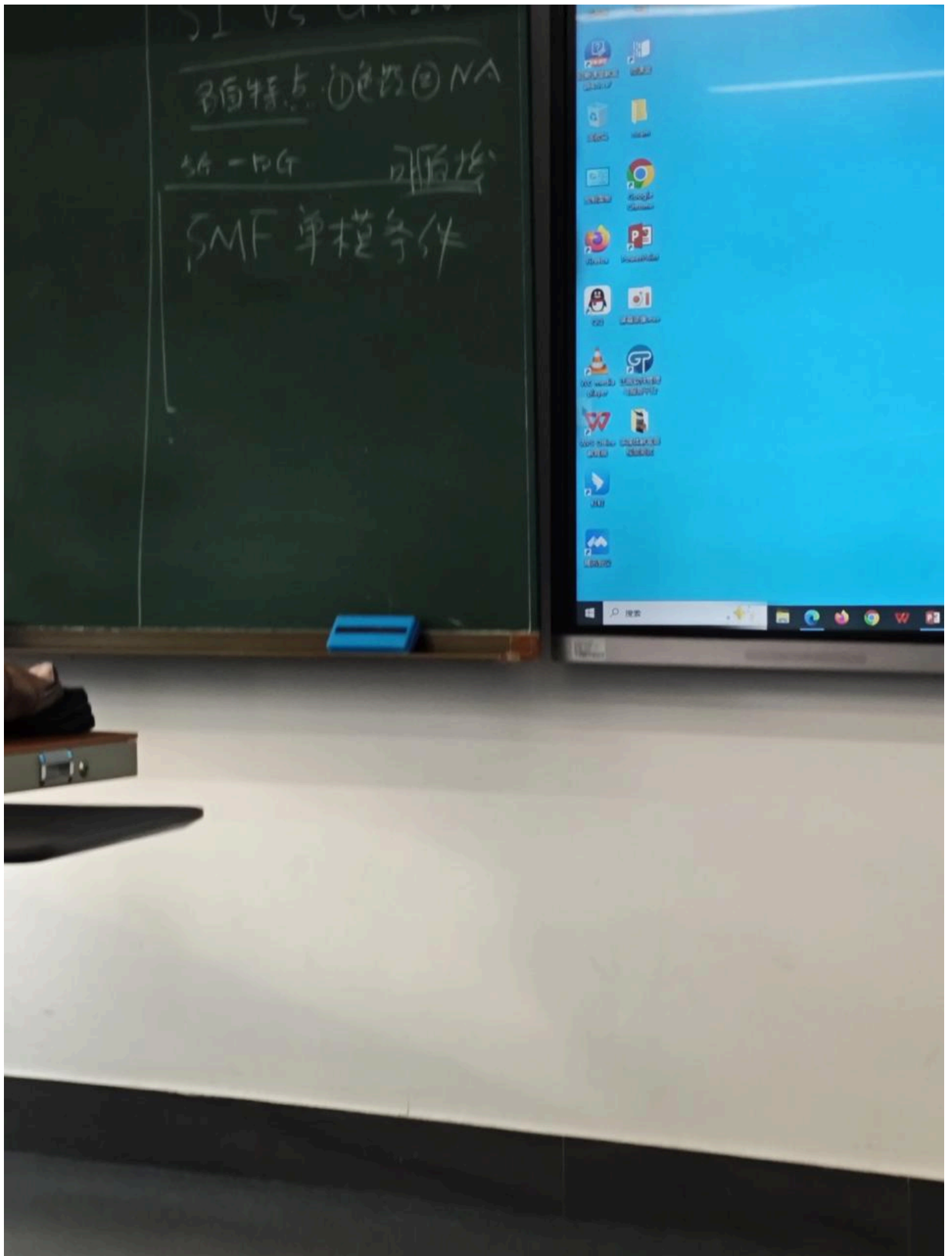


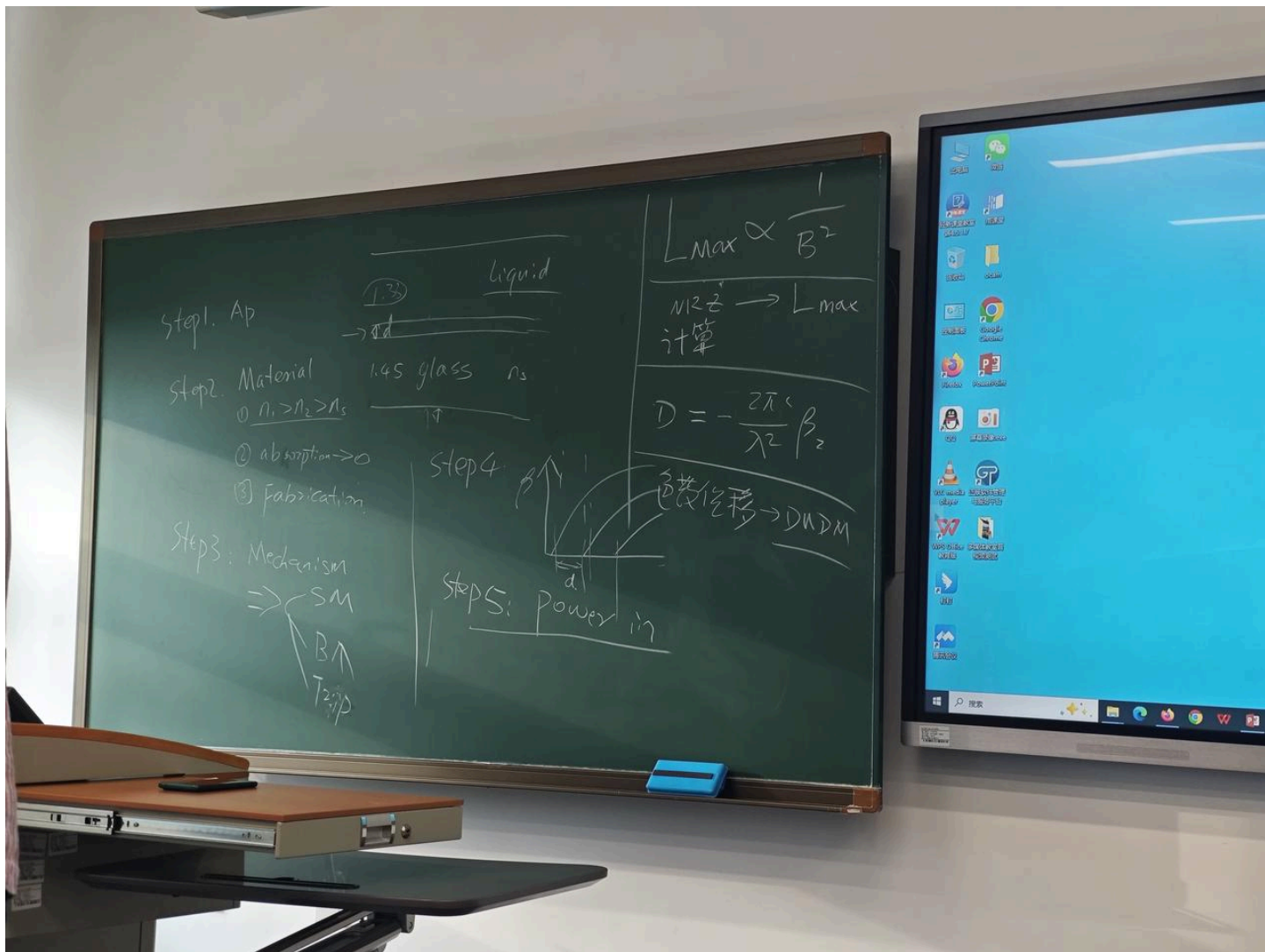
$$\phi(t) \propto V(t)$$

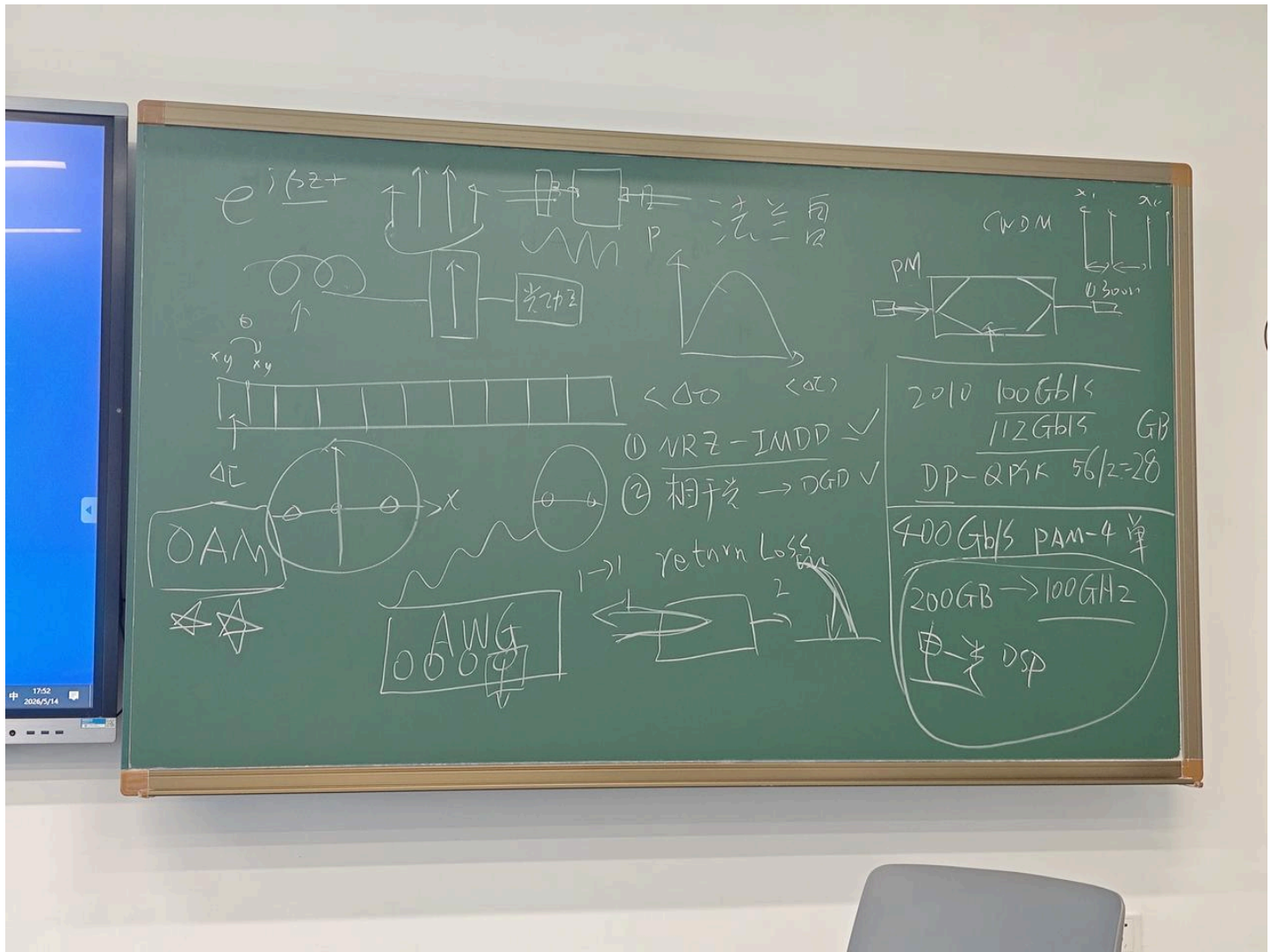
$$\frac{d\phi(t)}{dt} = \omega(t)$$

$V \rightarrow 0 \rightarrow 1 \rightarrow 0$
 $0 \rightarrow 180^\circ \rightarrow V_R \rightarrow -1 \rightarrow 180^\circ$

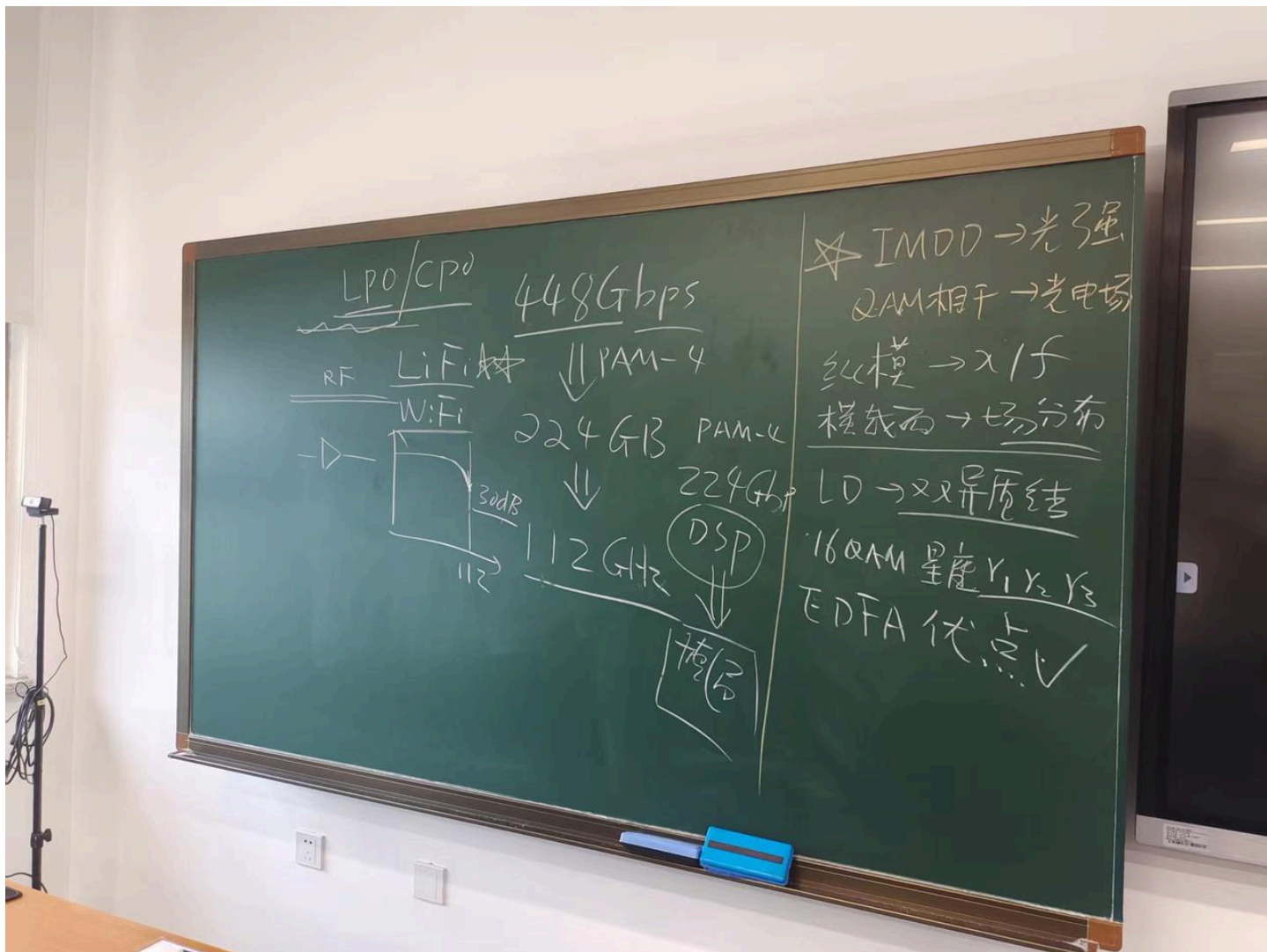








0519



$\left(\begin{array}{c} \text{2} \\ \text{dB} \end{array} \right)$ 指数
(通信测量)

光信噪比推导

ASE: 谱密度/单/双

OSNR $\longleftrightarrow$ SNR 关系式

物

ad

光

FEC

$X_I X_Q$

$Y_I Y_Q$

PRBS

Correlation

01001000

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PIN vs APD

相干光接收机
装置、原理、公式推导

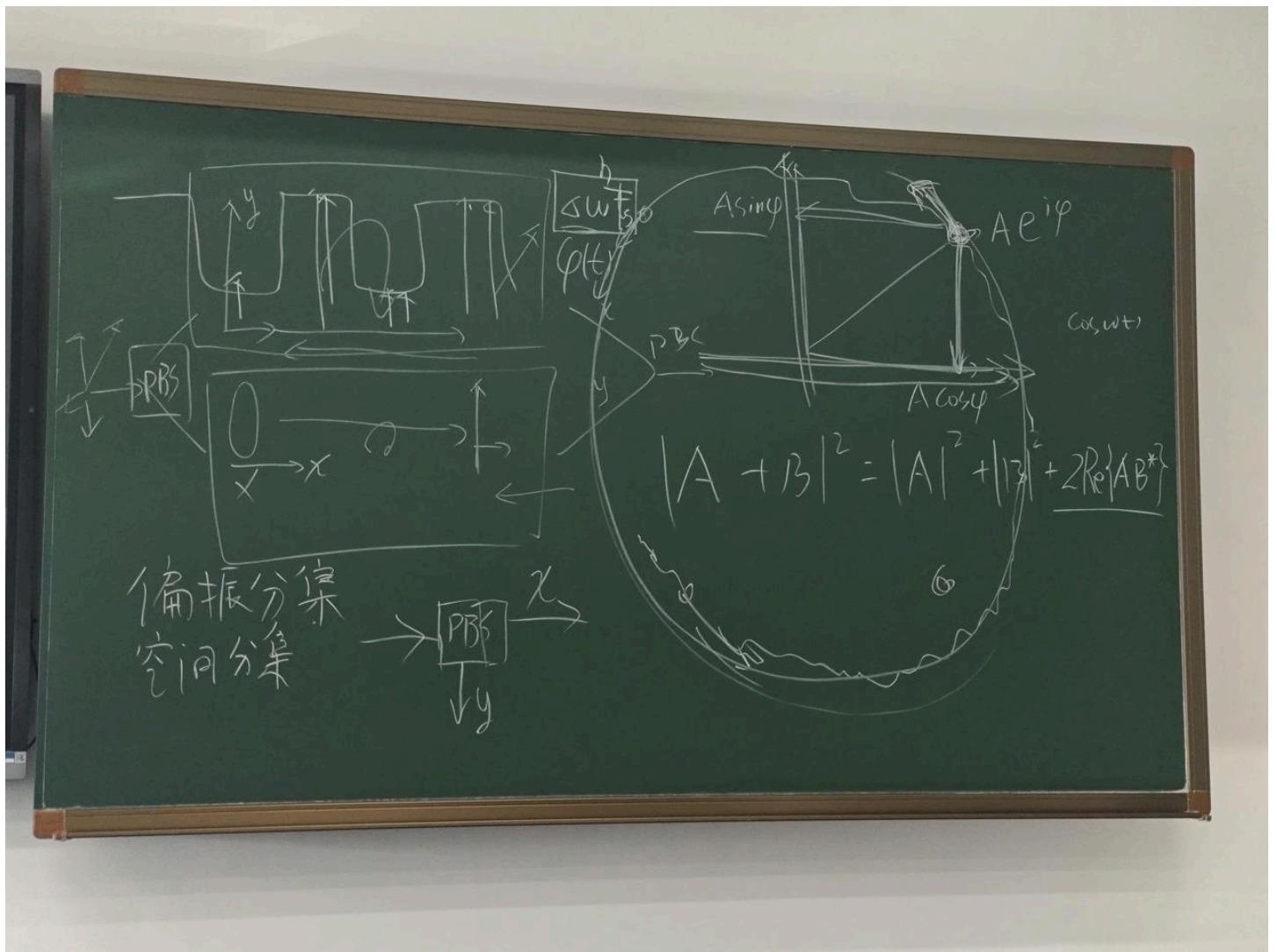
偏振 $\rightarrow \theta \rightarrow H, V$

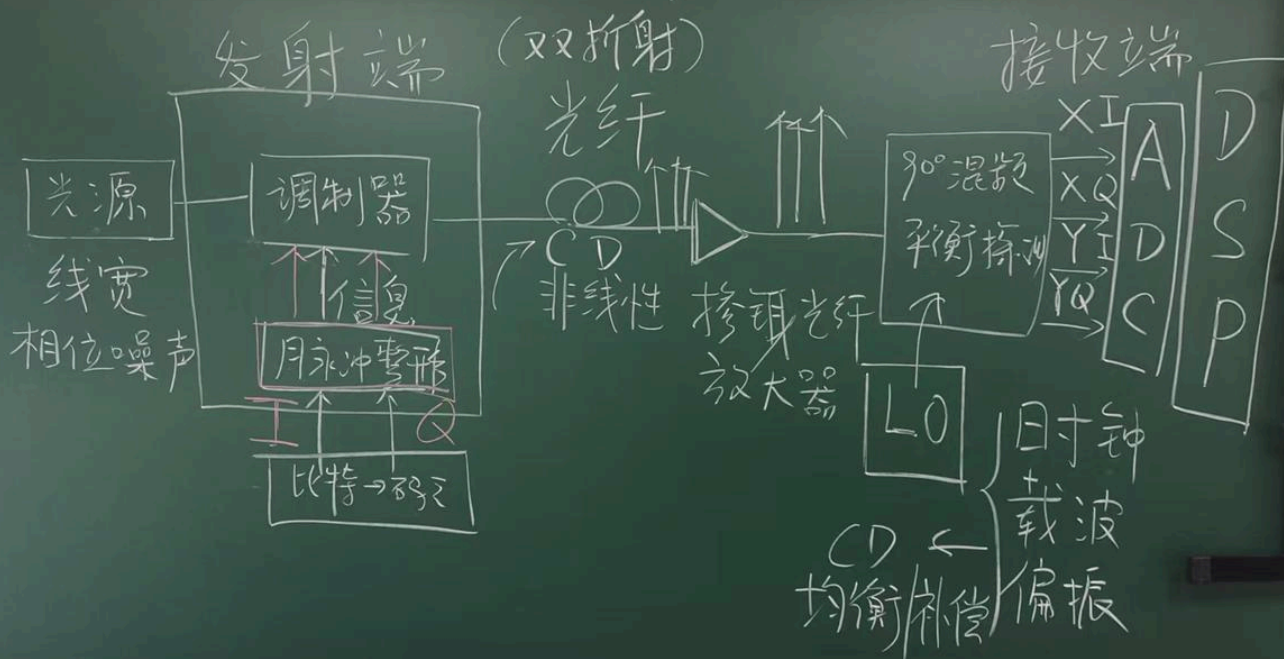
载波 $\cos(\omega_c t)$ $\rightarrow \phi$ 相位

时域时钟 $\rightarrow \omega_c t$

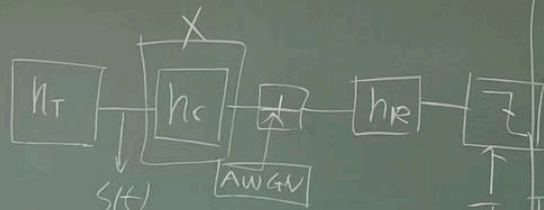
帧同步 DSP处理
流程

<https://w>

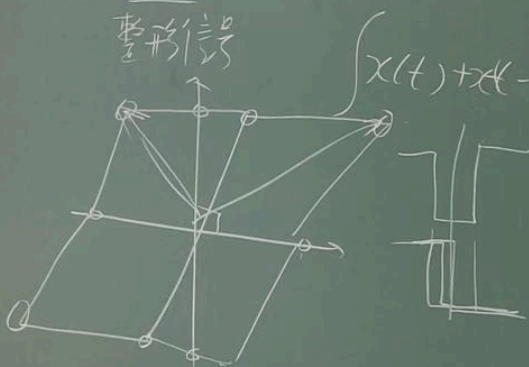




0100
码元



整形滤波

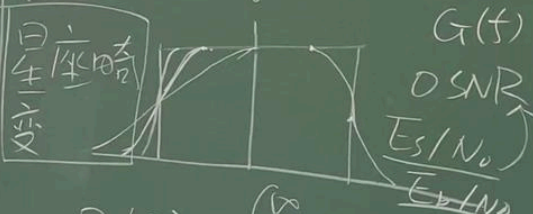


$x(t) + x(t - T)$

$$\frac{1}{T_s} \sum_{m=-\infty}^{\infty} G(f - \frac{m}{T_s}) = 1$$



星座图
变



$$R(0) = \int_{-\infty}^{\infty} P(f) df$$

维纳-辛钦定理



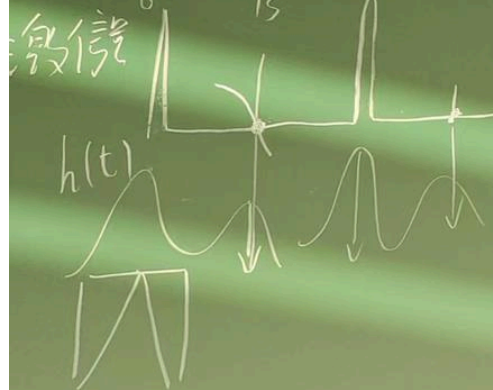
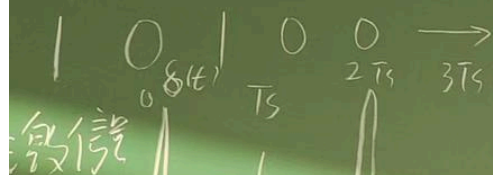
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无线 $\rightarrow$ QAM / QPSK

无线 $\rightarrow$ QAM / QPSK

$$\underline{1} \times 60 \text{ s wt}$$

$2 \times \sin \omega t$



$$\textcircled{\times} \frac{h(t)}{H(w)}$$

1) ISI-free $\max \frac{S}{N} \rightarrow$ 匹配滤波

2) clock / carrier / pol.
 Δt ω/ϕ θ

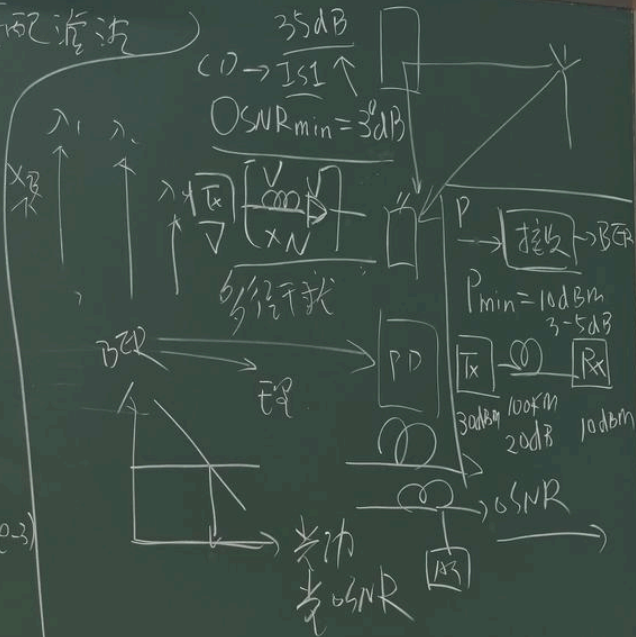
3) 偏振分集 X Y 轴

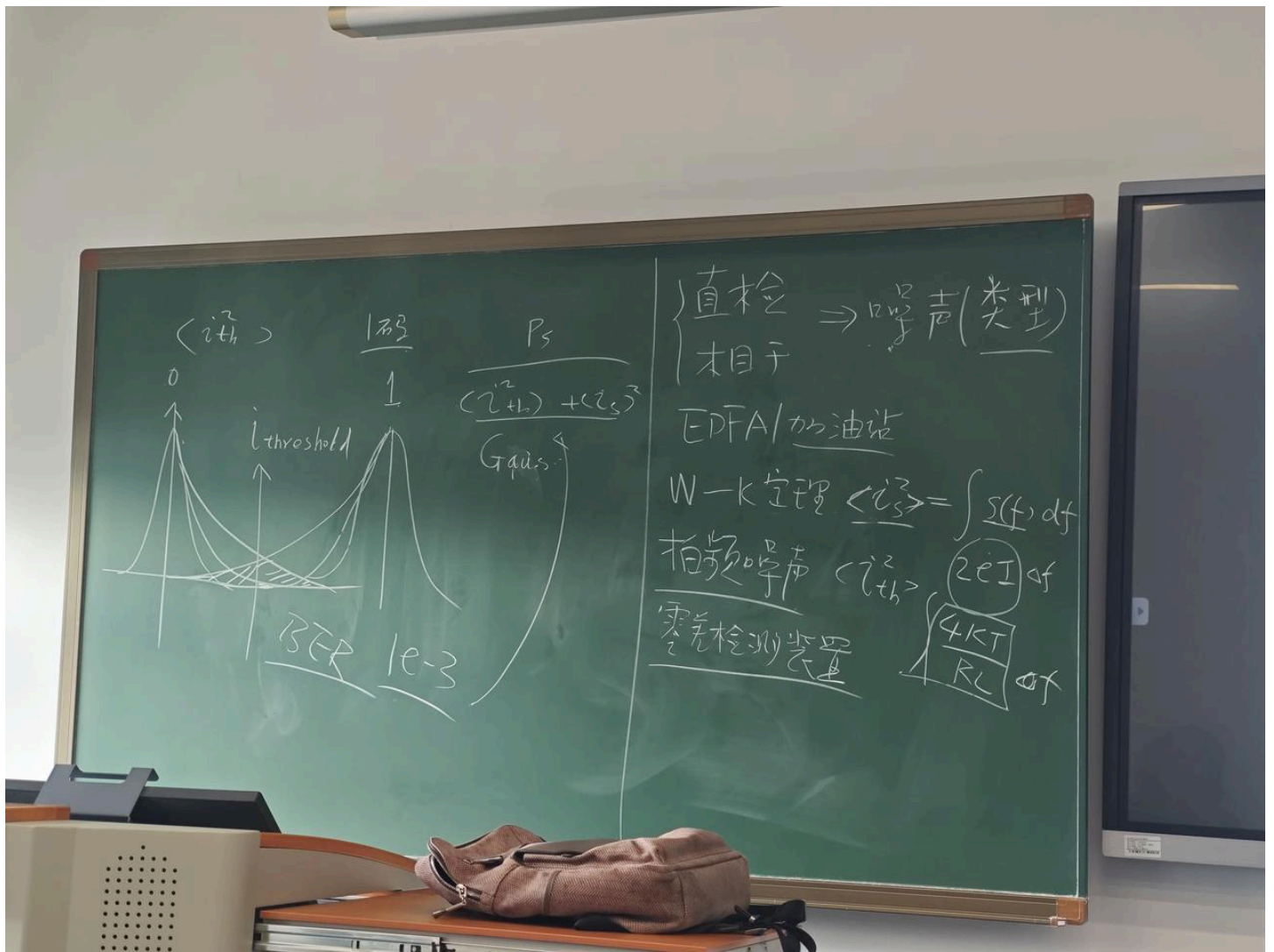
相位分集: $\cos(\omega t)$ $\sin(\omega t)$

零差/外差 $\downarrow$ 平衡/单端

光功率 vs 光信噪比

4) 噪声 $\rightarrow$ 分布 $\rightarrow$ 积分 $\rightarrow$ BER (10^{-3})
 代价 $\leftarrow$ 实际 $\leftarrow$ 精度





0604

★★★★★

$SNR_e \longleftrightarrow OSNR$

多段OSNR公式

$OSNR(dB) = P_{in} + \dots$

★★★★ 噪声 N.

OOK 热噪声 BER

★★ →

QAM-BER

接收机灵敏度 → 泊松分布

热噪声极限 SNR_e

散弹噪声极限 SNR_e

★ $SNR \longleftrightarrow P_s$

★ 哪种方案实现散弹 SNR

★ SNR vs M

★ $NEP \propto \sqrt{\Delta f}$ ★★
公式自推

EDFA 噪声指数推导 ★★

Changjiang 杨春雨 Start Slide Si
Server Login

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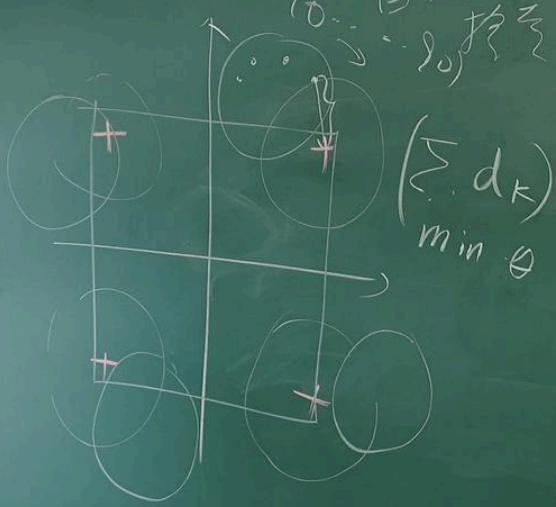
28

幻灯片第 26 张, 共 48 张

搜索

信号 + ASZ

盲相位
10... 20... 30...



QPSK BER公式应用

$$\frac{E_s}{N_b} \longleftrightarrow OSNR$$

两种场景 OSNR比较

★★★★ 58 + P_{in}

$$-k \log_{10}(x) - L(dB) - F$$

★★★★ $F_{sys} = F_1 + \frac{F_2 - 1}{G}$

$$\frac{P_{Tx}}{N} = OSNR \left(\log(OSNR) \right) - () dB$$

$$N * S_{ASE} \cdot B_{ref} \cdot 0.118 \quad (58 + P_{in} - F + 10 \log(W))$$

单偏振 x 2

$$(G-1) n_{sp} h \nu$$

不归并 DSP 功能
★★★ 载波恢复原理

